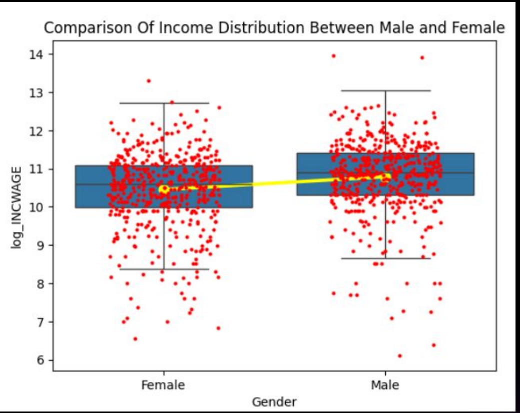
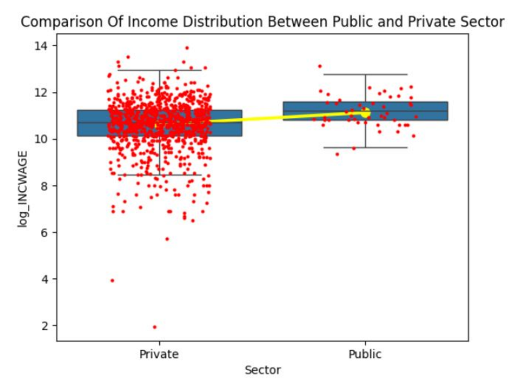
Does the gender wage gap change when we control for sector worked?

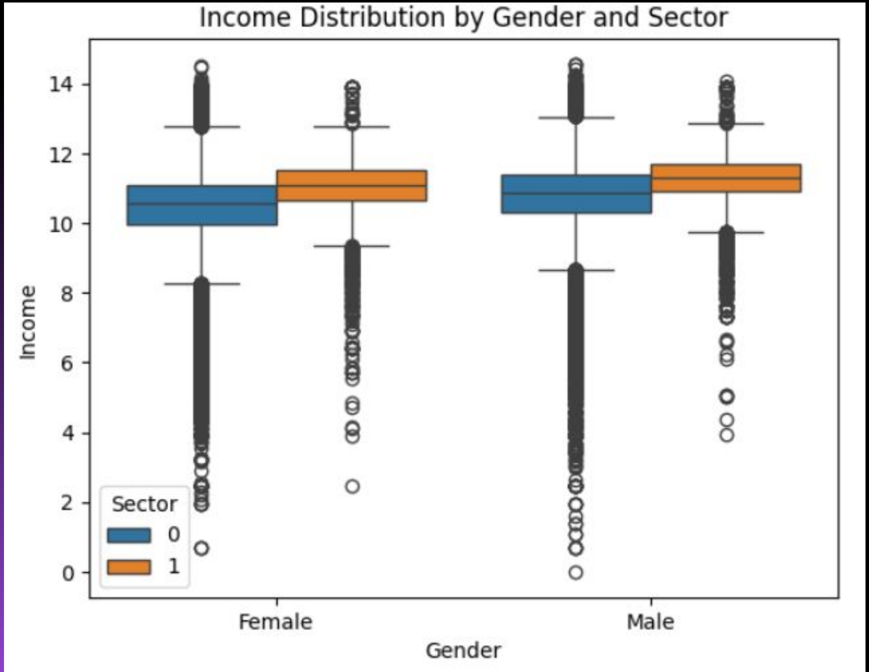


Gender Variables
0 = Female
1 = Male



Sector Variables
0 = Private
1 = Public/Gov

Wage = β0 + β1*Gender + β2*Sector



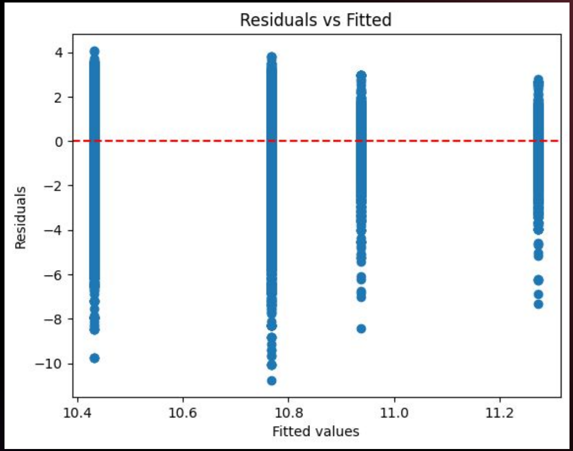
Residuals

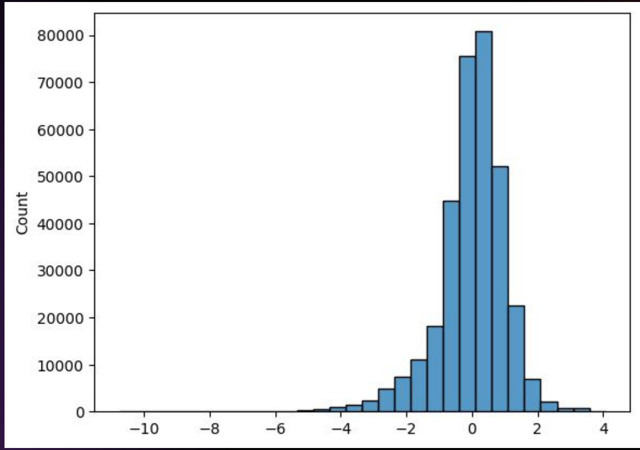
Uncontrolled for sector

```
model2= smf.ols('INCWAGE ~ SEX', data=data).fit()
print(model2.summary())

                            OLS Regression Results
==============================================================================
Dep. Variable:                INCWAGE   R-squared:                       0.016
Model:                            OLS   Adj. R-squared:                  0.016
Method:                 Least Squares   F-statistic:                     5557.
Date:                Mon, 27 Apr 2026   Prob (F-statistic):               0.00
Time:                        23:49:39   Log-Likelihood:            -4.4267e+06
No. Observations:              347879   AIC:                         8.853e+06
Df Residuals:                  347877   BIC:                         8.853e+06
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   5.199e+04    200.614    259.165      0.000    5.16e+04    5.24e+04
SEX         2.058e+04    276.086     74.543      0.000       2e+04    2.11e+04
==============================================================================
```

Controlled for sector

```
model1 = smf.ols('INCWAGE ~ CLASSWKR + SEX', data=data).fit()
print(model1.summary())

                            OLS Regression Results
==============================================================================
Dep. Variable:                INCWAGE   R-squared:                       0.019
Model:                            OLS   Adj. R-squared:                  0.019
Method:                 Least Squares   F-statistic:                     3307.
Date:                Thu, 23 Apr 2026   Prob (F-statistic):               0.00
Time:                        19:05:11   Log-Likelihood:            -4.4261e+06
No. Observations:              347879   AIC:                         8.852e+06
Df Residuals:                  347876   BIC:                         8.852e+06
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept   5.116e+04    201.957    253.331      0.000    5.08e+04    5.16e+04
CLASSWKR    2.345e+04    726.760     32.272      0.000      2.2e+04   2.49e+04
SEX          2.05e+04    275.686     74.350      0.000       2e+04     2.1e+04
==============================================================================
```





Results:

- Males still earn about 40% more than females even after controlling for sector. This means that the sector of employment does not work to explain the wage gap.

Residuals:

- The model is appropriate but there is some left skewness and mild heteroskedasticity. This reaffirms that factors beyond the sector are driving wage differences between men and women.